prime239v1 has a low L(1, χ): the early Euler factors are the L-value

2026-10-30 · `research/nist_seed_lvalue_20261030/`

Status: complete. Protocol `PROTOCOL.md` committed and pushed (`dd218eb5`) before any study statistic was computed. One deviation (D1): the first launch was killed by a harness time limit before writing results, and the identical command was rerun. Analysis only: no ECDLP cost changes, and no hypothesis status changes.

Answer

`papers/nist-seed-split-null/paper.md` §6 left open whether prime239v1's low split count reflects a small L(1, χ_{D_K}), or only low early Euler factors with an ordinary L-value. **The L-value is low.**

prime239v1	value
f_π, D_K	1, D_K = Δ, |D_K| ≈ 2^240.9
$\hat{L}(1, \chi_{D_K})$, Euler product to x = 10^8	**0.2461** (log $\hat{L}$ = −1.4019)
percentile under prime-order null	**0.25%** (second seed 0.23%)
rank of log $\hat{L}$ among the 61 curves	**1 of 61** (lowest)
protocol decision (Q1)	**low** (< 1st percentile)
implied class number h = √|D_K|·L/π	≈ 2^116.8, against ≈ 2^117.6 at the null median

Where the L-value comes from. log $\hat{L}$ = −1.402 splits as:

- $\ell = 2$: −0.405, the same for every odd prime-order curve here, since Δ ≡ 5 (mod 8);
- $3 \le \ell \le 691$: −0.998;
- $\ell > 691$: +0.002.

So the first 124 odd primes carry essentially all of the deviation. The weighted statistic E(B) = $-\Sigma_{3\le\ell\le B} \log(1 - \chi(\ell)/\ell)$ has percentiles of 1.0 at B = 101, 0.69 at 251, 0.36 at 397, 0.24 at 691, 0.22 at 997, 0.24 at 2003, and 0.25 at 5003.

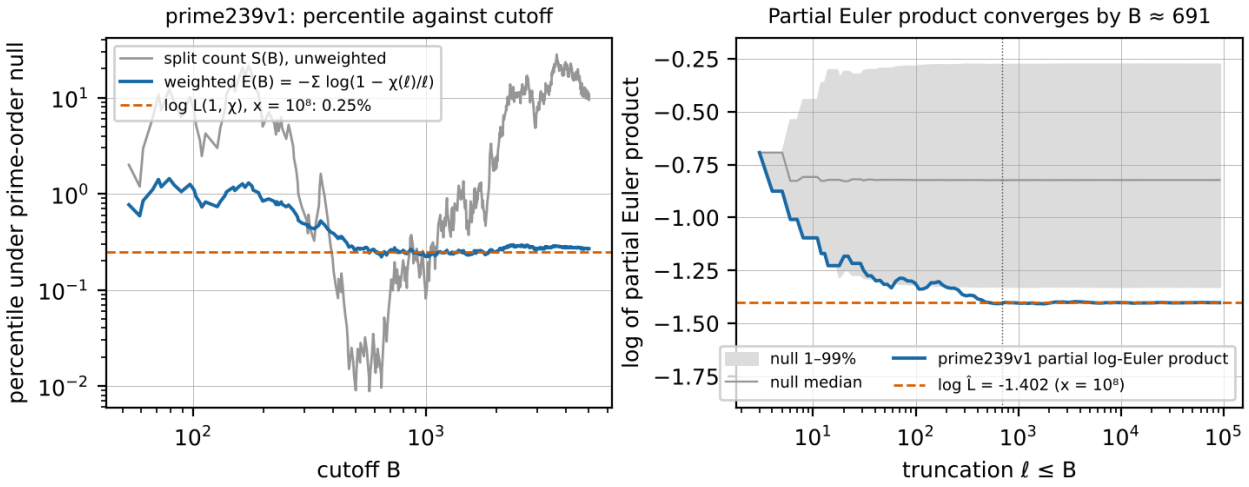


Figure 1: prime239v1: percentile against cutoff, and partial Euler product against truncation

Left: the unweighted split count's percentile (grey, from the split-null study's 655-cutoff sweep) moves from 0.009% to about 20% as B grows. The weighted E(B) (blue) settles at the L-value's percentile (dashed) by B ≈ 500 and stays there. Right: the partial log-Euler product leaves the null 1–99% band before B = 100 and has converged to log $\hat{L}$ by B ≈ 691.

What this corrects

The paper's "possible explanation, not established" was wrong: prime239v1's L-value is not ordinary. Between B = 691 and 5003 its *split-count* percentile rises from 0.06 to 11. That rise is an artefact of

counting primes without weight. The added primes near B contribute about $1/\ell$ each to log L, so they cannot move it. Yet each of them shifts the unweighted count by a full unit.

What this does not show

- **No new independent evidence.** prime239v1 was picked out because its S was extreme. Across the 61 curves, corr(E(691), log $\hat{L}$) = 0.95 and corr(S(691), log $\hat{L}$) = 0.46. The low L-value restates the same anomaly in a weighting that does not fade. If nothing had been selected, the chance that some curve among 61 reaches the 0.25th percentile is $1 - (1 - 0.0025)^{61} \approx 0.14$. Among the 8 published curves it is about 0.02.
- **No published/unpublished shift.** The median log-L percentiles are 39 for the 8 published curves, 49 for the 11 named omitted curves, and 52 for the 42 held-out curves.
- **No CM-method curve.** $|D_K| \approx 2^{241}$, and h is within a factor of about 2 of its null median.
- **No ECDLP consequence.** No known ECDLP attack, generic or structural, depends on $L(1, \chi)$ or on the class number of the CM order. Within a level of the isogeny volcano, JMV random self-reducibility makes the problem uniformly hard. The atlas leads `LEAD-STR-jmv-random-self-reduction` and `LEAD-STR-torsion-module-leak` are unaffected.
- **Conditional on GRH.** $\hat{L}$ is a truncated Euler product. Under GRH the truncation error is expected to be of order 10^{-3}; this is not a proved bound.

Controls (all passed)

control	result
estimator against exact class numbers (20 random fundamental D, 80–110 bits)	max $\lvert\log \hat{L} - \log L\rvert = 4.0 \cdot 10^{-5}$ at $x = 10^8$ ($6.7 \cdot 10^{-4}$ at 10^6); threshold 0.01
χ: S(691) recomputed for all 61 curves against the committed held-out run	0 mismatches
A/A implementation: vectorized χ against Python `pow`, $\ell \le 10^6$, 3 curves × (Δ, D_K)	0 mismatches
D_K and f_π against the exact conductor census	17 of 17 match
A/A Monte Carlo (seeds 20261030, 20261031)	log L 0.247 / 0.226%; E(691) 0.238 / 0.213%
closed-form null law against `analysis_lib.probs`, $\ell < 400$, four fields	exact
convergence, every curve	max $\lvert\log \hat{L}(10^8) - \log \hat{L}(10^7)\rvert = 2.3 \cdot 10^{-4}$
null calibration: 42 held-out log-L percentiles against uniform	KS D = 0.066 (5% critical value 0.210)

Method

- **Curves:** the 61 curves of the held-out test (19 named and 42 held-out, in four fields), loaded from the same three inputs.
- **D_K:** D_K = `coredisc(Δ)` from PARI 2.17.2. Exact class numbers were infeasible here: `quadclassunit` takes 38 s at 140 bits and grows about 6× per 20 bits, so roughly a day per 241-bit discriminant.
- **Estimator:** $\log \hat{L} = -\Sigma_{\ell \le 10^8} \log(1 - \chi_{D_K}(\ell)/\ell)$, with χ computed vectorized (16-bit Horner residues, then Euler's criterion in int64).
- **Null:** Frobenius uniform in $GL_2(F_\ell)$ with determinant p and no eigenvalue 1, independent across ℓ, in closed form. 200,000 draws over $3 \le \ell \le 10^5$ per field; the $\ell > 10^5$ tail has variance below 10^{-6}.
- **Runtime:** 10 m 43 s wall on 4 vCPUs.
- **Provenance:** `runs/commit.txt` and `runs/env.txt`.

Reproduce

```
cd research/nist_seed_lvalue_20261030
python3 lvalue.py ../nist_seed_1997_selection_extension_20261023/sea_signature/runs/
scan-20261023T1840Z/raw.json \
  ../nist_seed_1997_acceptance_20261024/analysis.json \
```

```
  ../nist_seed_selection_20261016/raw/full_prime_order_certificates.jsonl \
  ../nist_seed_exact_conductor_census_20261023/runs NEW_OUT_DIR
python3 figures.py
```

Outputs: `runs/lvalues.json`, `runs/controls.json` and `runs/rows.jsonl`.